

GRASSWORKS: Analysis of Bauer et al. (submitted) Functional traits of grasslands: Community weighted mean of seed mass per plot (esy4 = 4qm plots)

MB

2026-05-28

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NOTE: To compare different models, you only have to change the models in the section ‘Load models’

Preparation

Protocol of data exploration (Steps 1-8) used from Zuur et al. (2010) Methods Ecol Evol DOI: 10.1111/2041-210X.12577

```
library(here)
library(tidyverse)
library(ggbeeswarm)
library(patchwork)
```

```
library(DHARMA)
library(emmeans)
```

Packages

```
sites <- read_csv(
  here("data", "processed", "data_processed_sites_esy4.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(
    .default = "?",
    eco.id = "f",
    region = col_factor(levels = c("north", "centre", "south"), ordered = TRUE),
    site.type = col_factor(
      levels = c("positive", "restored", "negative"), ordered = TRUE
    ),
    obs.year = "f",
    hydrology = "f",
    mngm.type = "f"
  )
) %>%
mutate(
  esy4 = fct_relevel(esy4, "R22", "R1A"),
  eco.id = factor(eco.id)
) %>%
rename(y = cwm.abu.seedmass)
```

Load data

Statistics

Data exploration

Means and deviations

```
Rmisc::CI(sites$y, ci = .95)
```

```
##      upper      mean      lower
## 0.001621297 0.001509416 0.001397534
```

```
median(sites$y)
```

```
## [1] 0.00126
```

```
sd(sites$y)
```

```
## [1] 0.0009697053
```

```
quantile(sites$y, probs = c(0.05, 0.95), na.rm = TRUE)
```

```
##      5%      95%
## 0.000375 0.003310
```

```
sites %>% count(eco.id)
```

```
## # A tibble: 3 x 2
##   eco.id     n
##   <fct> <int>
```

```
## 1 654      101
## 2 664      91
## 3 686      99
```

```
sites %>% count(site.type)
```

```
## # A tibble: 3 x 2
##   site.type      n
##   <ord>      <int>
## 1 positive     45
## 2 restored    221
## 3 negative     25
```

```
sites %>% count(esy4)
```

```
## # A tibble: 2 x 2
##   esy4      n
##   <fct> <int>
## 1 R22    210
## 2 R1A     81
```

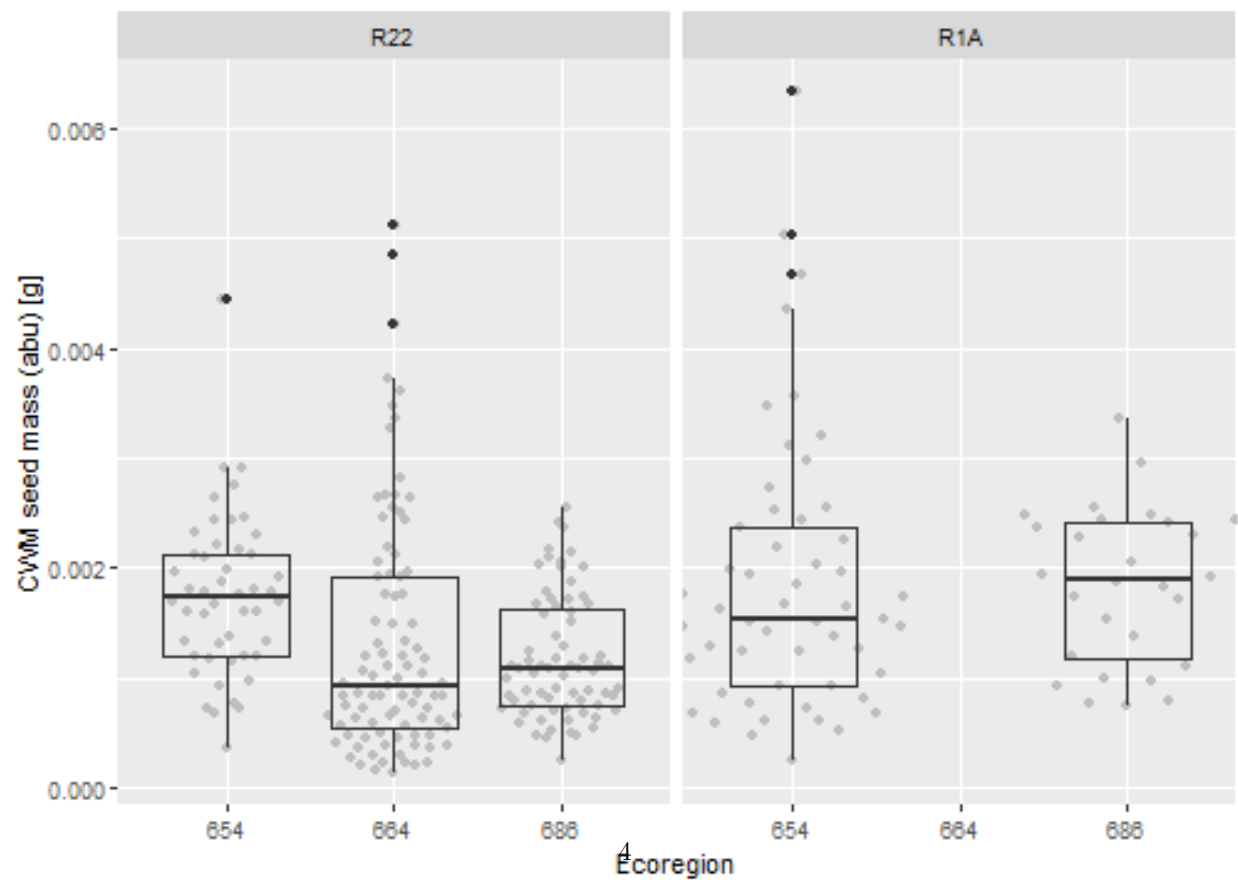
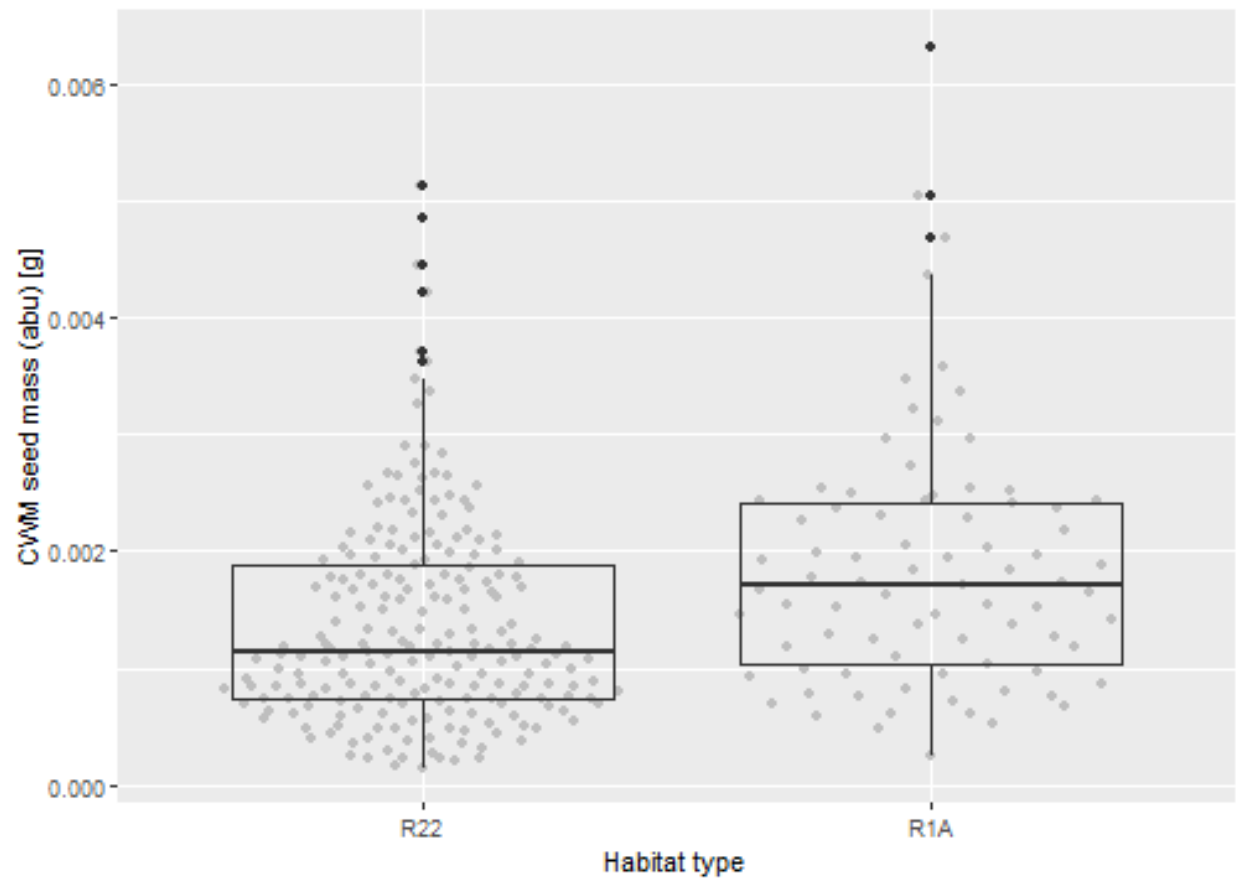
```
sites %>% count(esy4, eco.id)
```

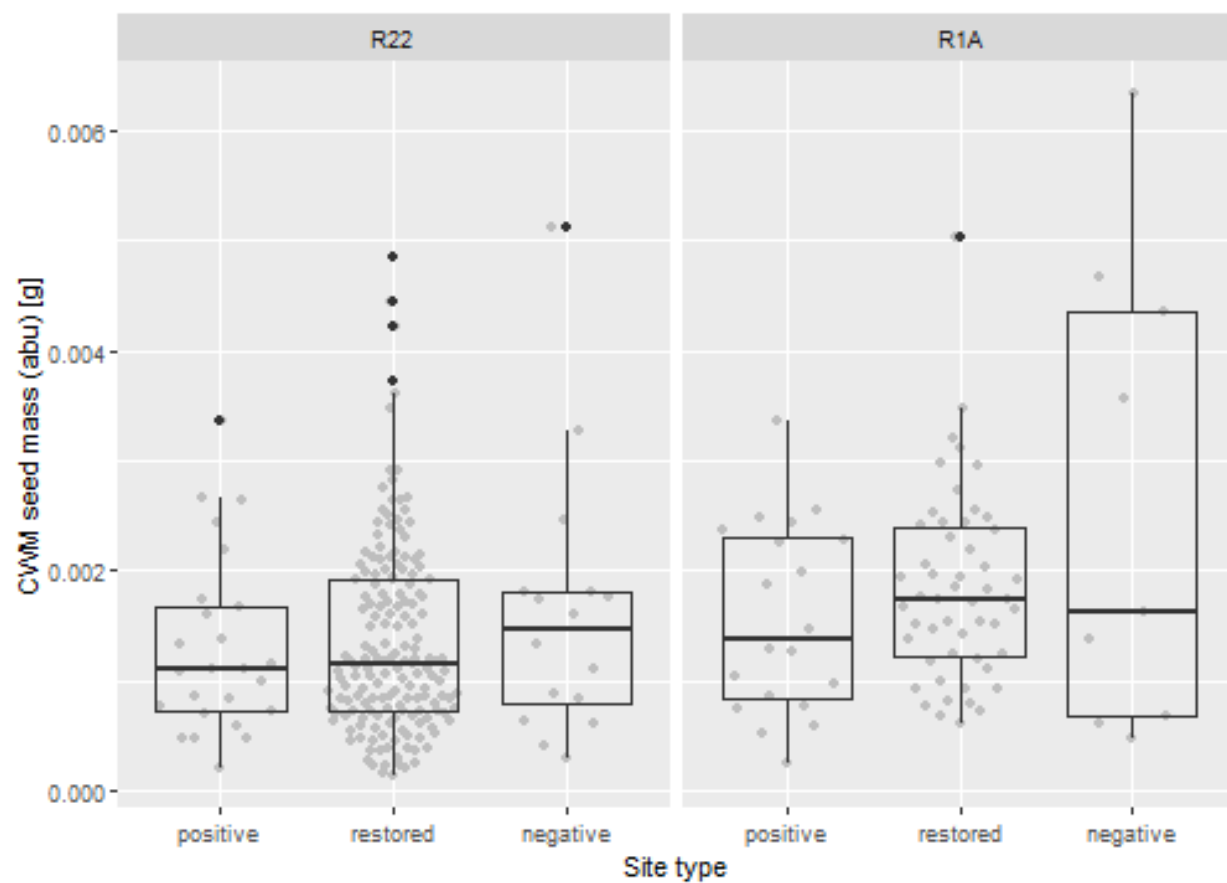
```
## # A tibble: 5 x 3
##   esy4 eco.id      n
##   <fct> <fct> <int>
## 1 R22   654     48
## 2 R22   664     91
## 3 R22   686     71
## 4 R1A   654     53
## 5 R1A   686     28
```

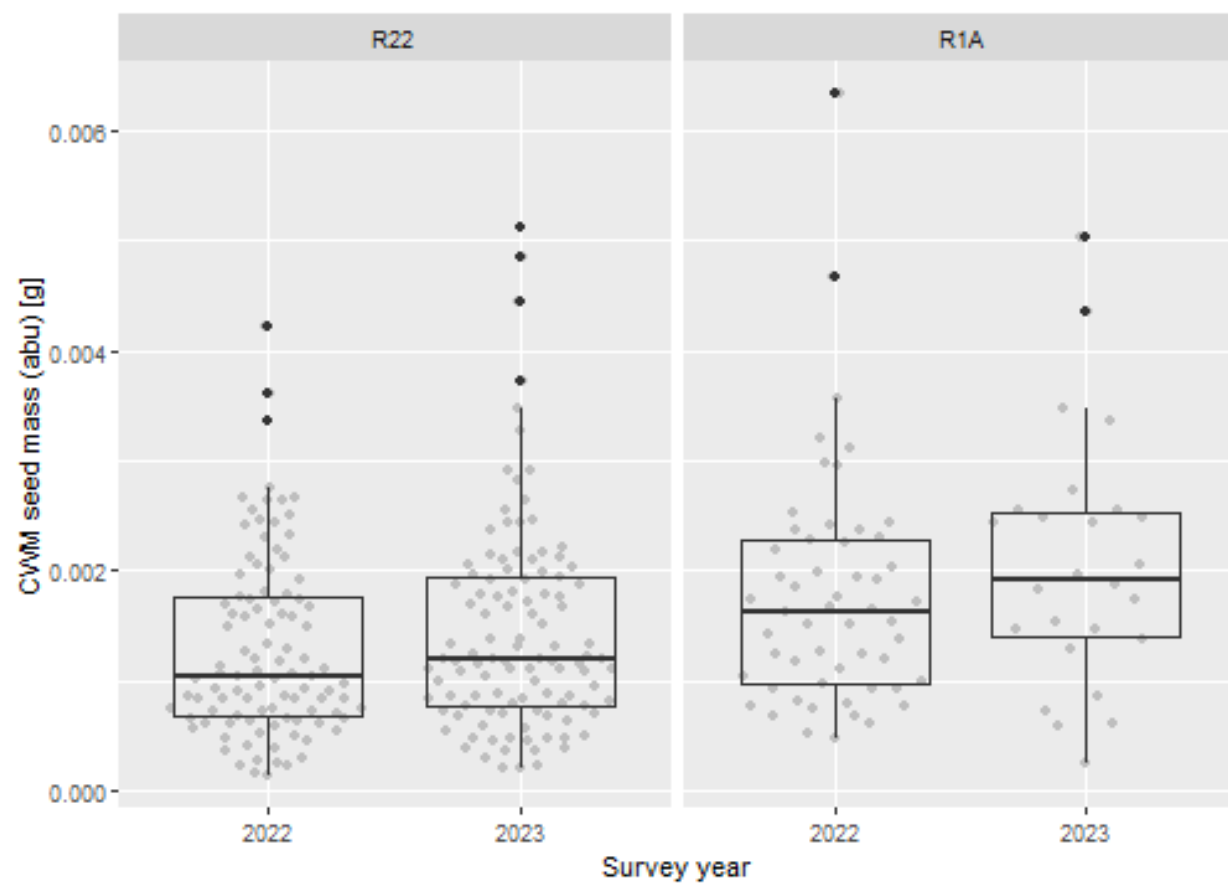
```
sites %>% count(esy4, site.type)
```

```
## # A tibble: 6 x 3
##   esy4 site.type      n
##   <fct> <ord>      <int>
## 1 R22   positive     25
## 2 R22   restored    169
## 3 R22   negative     16
## 4 R1A   positive     20
## 5 R1A   restored     52
## 6 R1A   negative      9
```

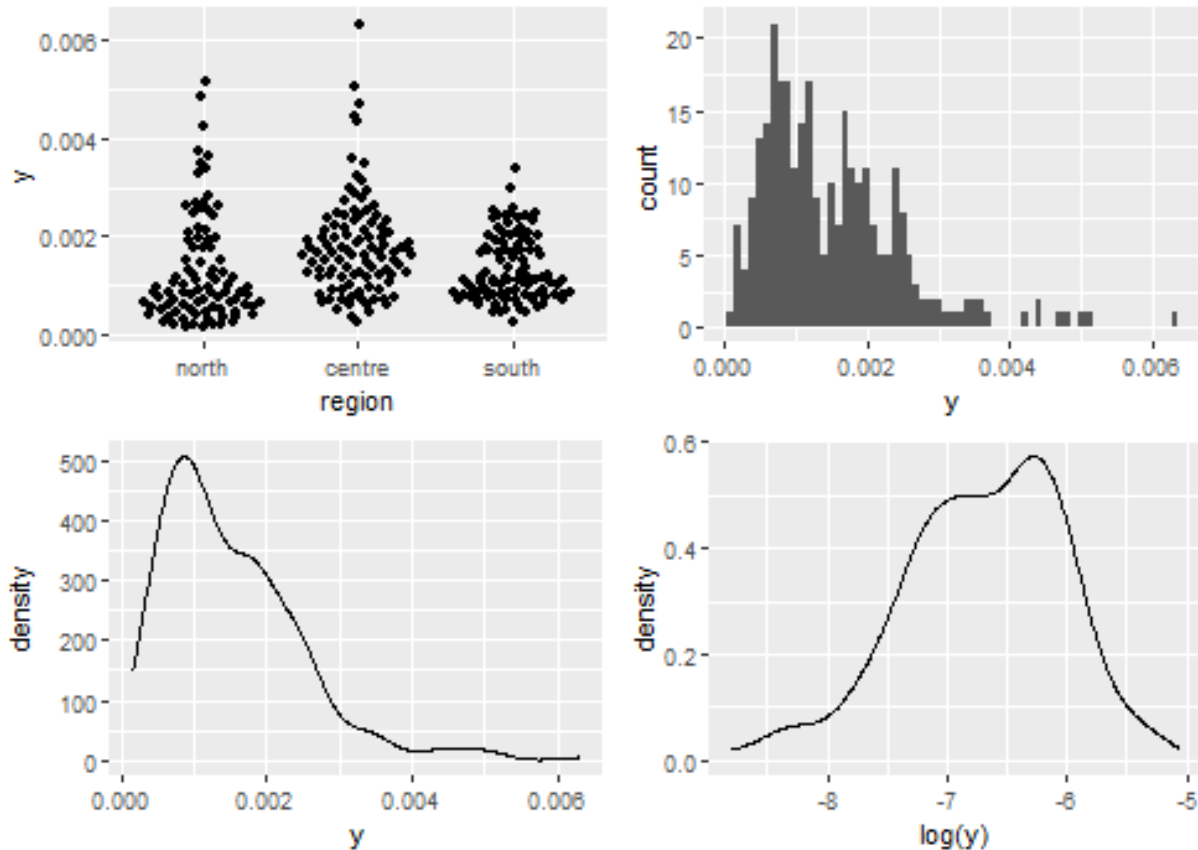
Graphs of raw data (Step 2, 6, 7)







Outliers, zero-inflation, transformations? (Step 1, 3, 4)



Check collinearity part 1 (Step 5)

Exclude $r > 0.7$ Dormann et al. 2013 *Ecography* DOI: 10.1111/j.1600-0587.2012.07348.x

```
# sites %>%  
#   select(where(is.numeric), -y, -starts_with("cum.")) %>%  
#   GGally::ggpairs(  
#     lower = list(continuous = "smooth_loess")  
#   ) +  
#   theme(strip.text = element_text(size = 7))  
  
# -> no continuous variables
```

Models

NOTE: Only here you have to modify the script to compare other models

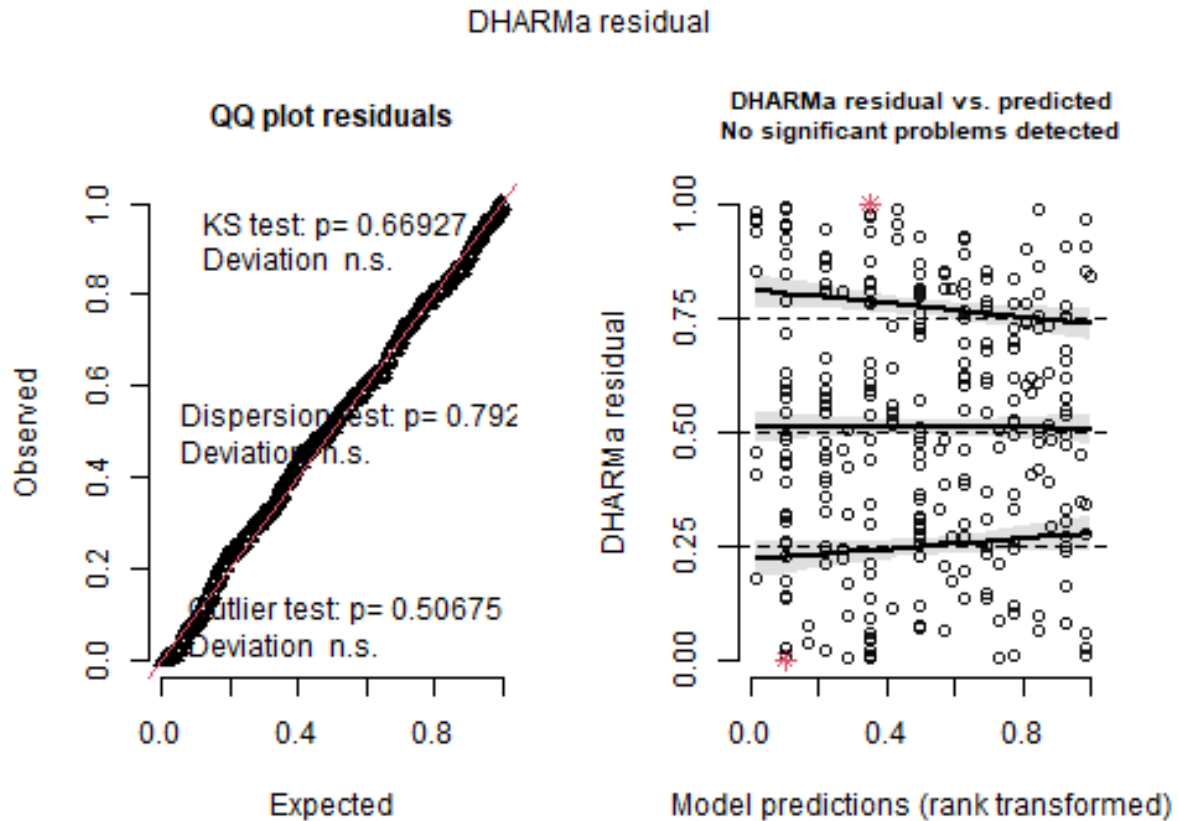
```
load(file = here("outputs", "models", "model_seedmass_esy4_cwm_1.Rdata"))  
load(file = here("outputs", "models", "model_seedmass_esy4_cwm_2.Rdata"))  
m_1 <- m1  
m_2 <- m2  
  
m_1@call  
## lmer(formula = log(y) ~ esy4 * (site.type + eco.id) + obs.year +  
##       (1 | id.site), data = sites, REML = FALSE)
```

```
m_2@call
## lmer(formula = log(y) ~ (esy4 + site.type + eco.id + obs.year)^2 +
##       (1 | id.site), data = sites, REML = FALSE)
```

Model check

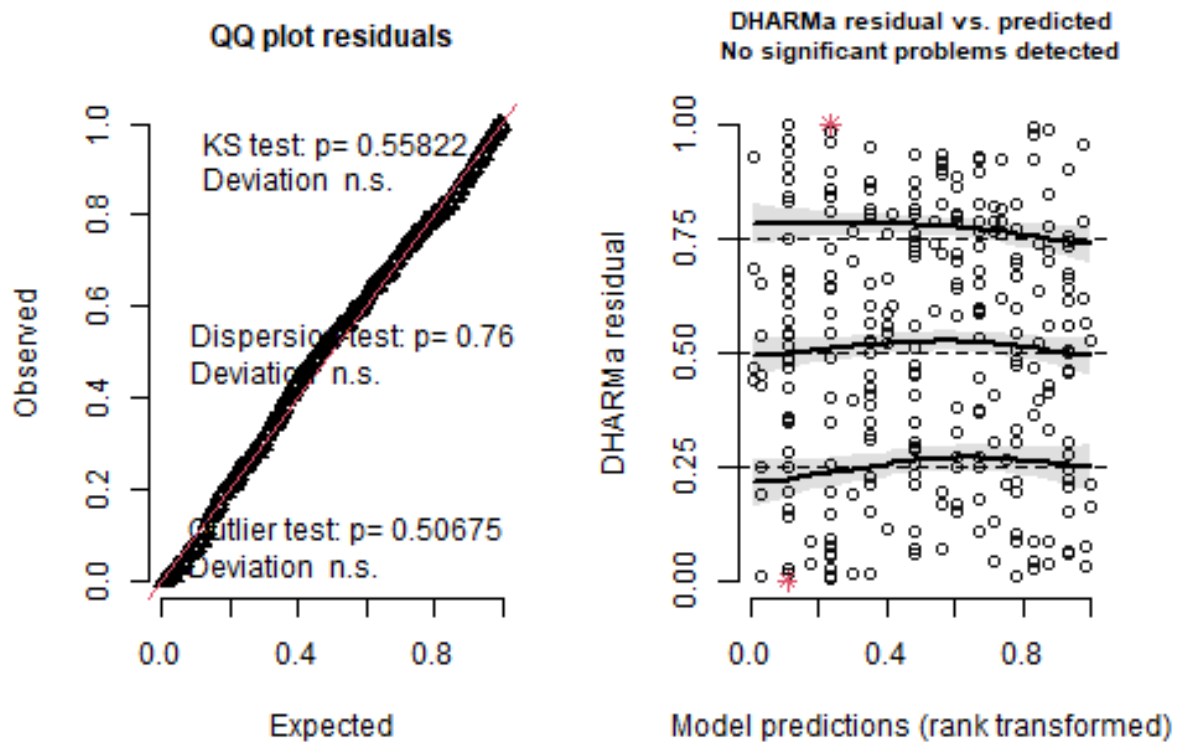
DHARMA

```
simulation_output_1 <- simulateResiduals(m_1, plot = TRUE)
```

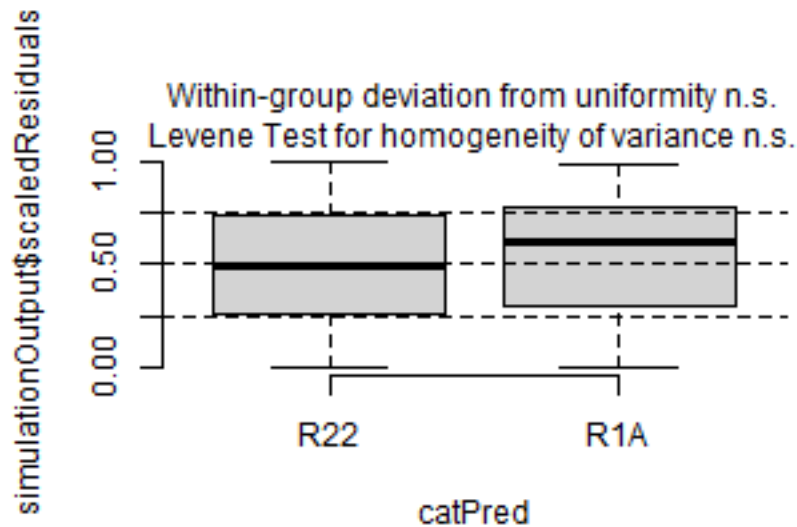


```
simulation_output_2 <- simulateResiduals(m_2, plot = TRUE)
```

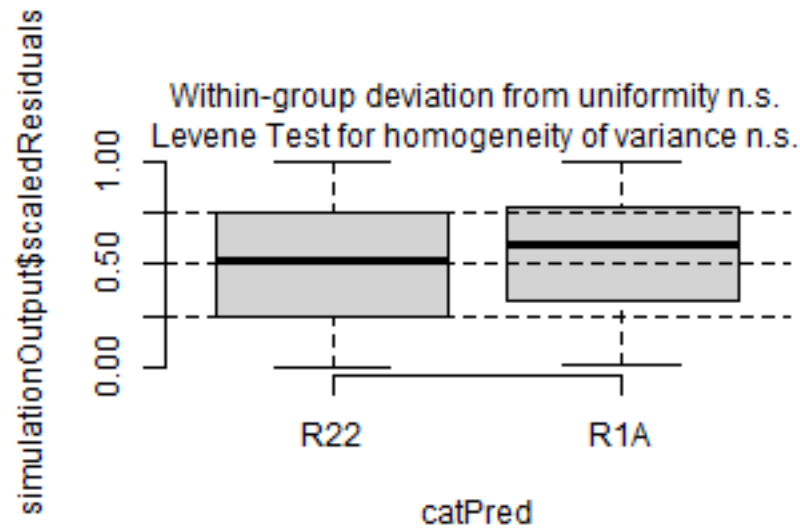

DHARMa residual



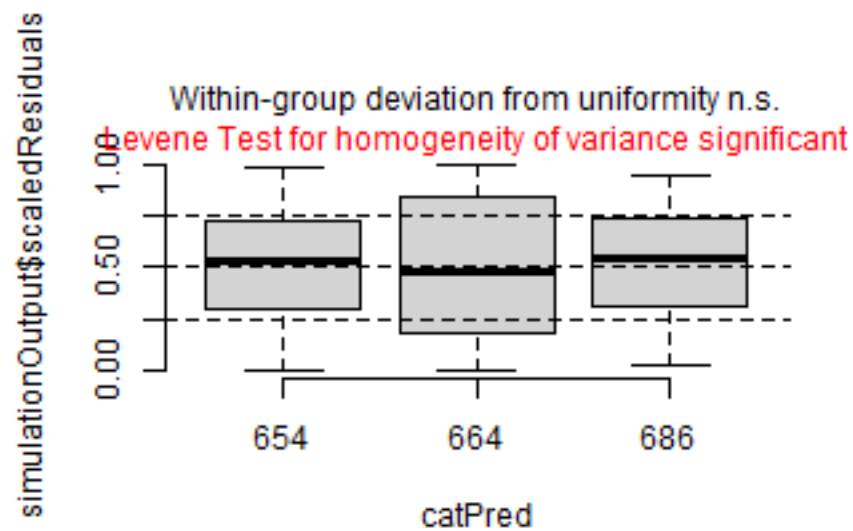
```
plotResiduals(simulation_output_1$scaledResiduals, sites$esy4)
```



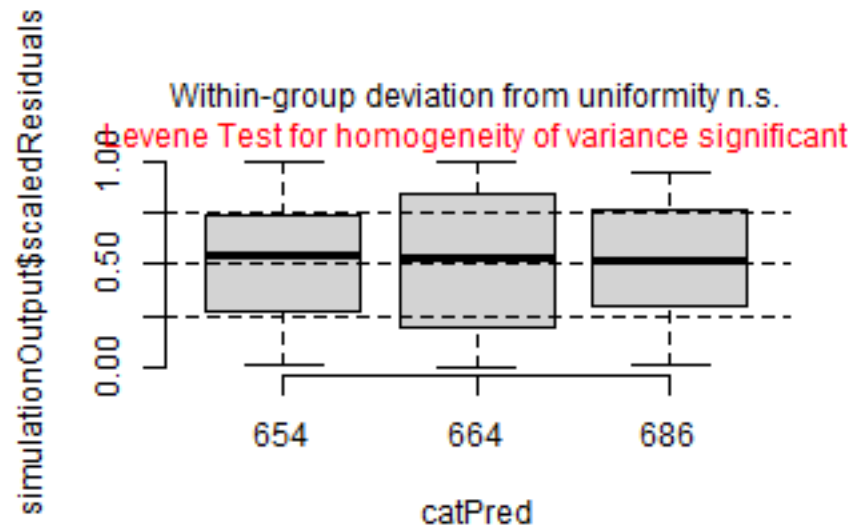
```
plotResiduals(simulation_output_2$scaledResiduals, sites$esy4)
```



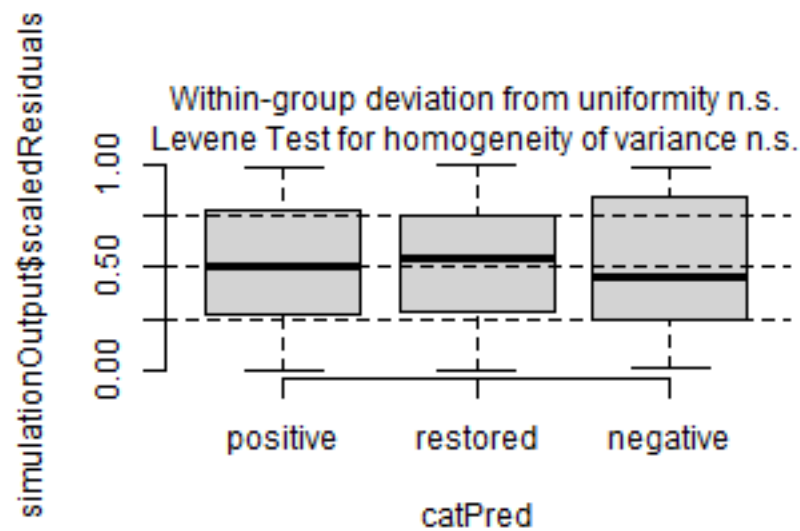
```
plotResiduals(simulation_output_1$scaledResiduals, sites$eco.id)
```



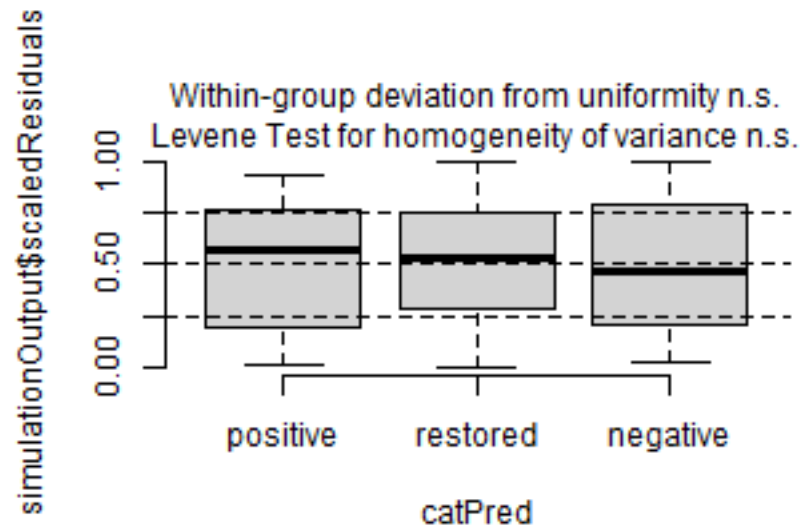
```
plotResiduals(simulation_output_2$scaledResiduals, sites$eco.id)
```



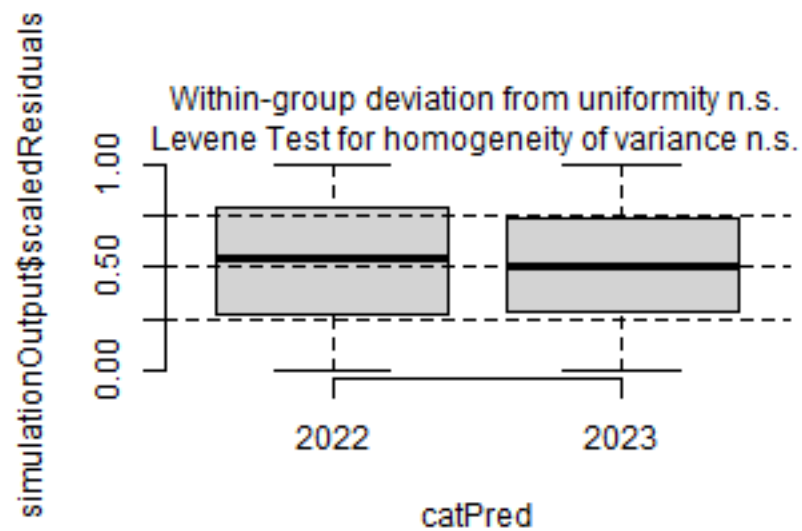
```
plotResiduals(simulation_output_1$scaledResiduals, sites$site.type)
```



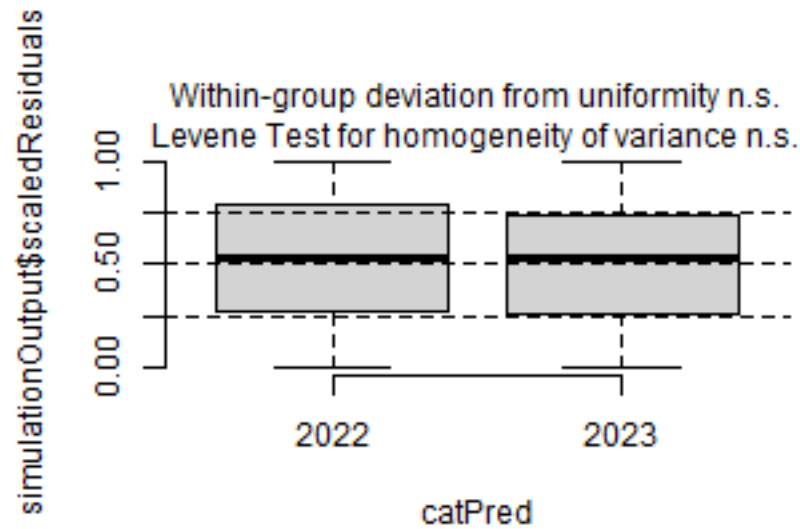
```
plotResiduals(simulation_output_2$scaledResiduals, sites$site.type)
```



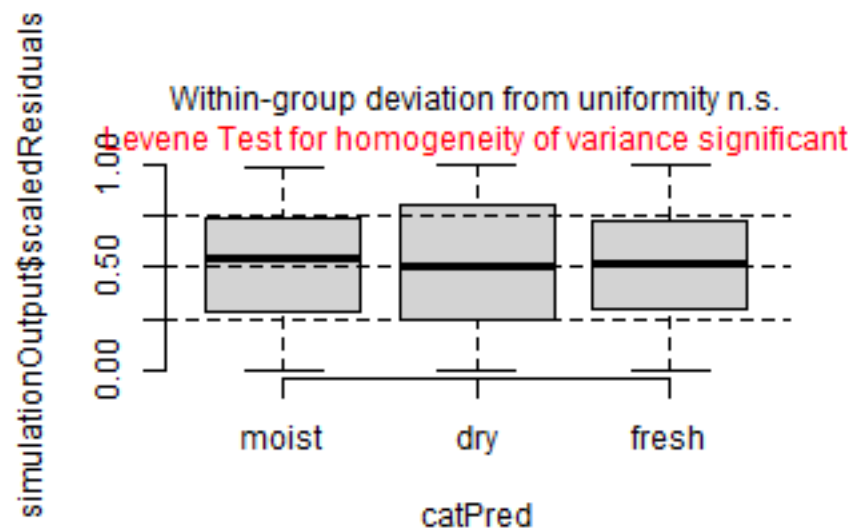
```
plotResiduals(simulation_output_1$scaledResiduals, sites$obs.year)
```



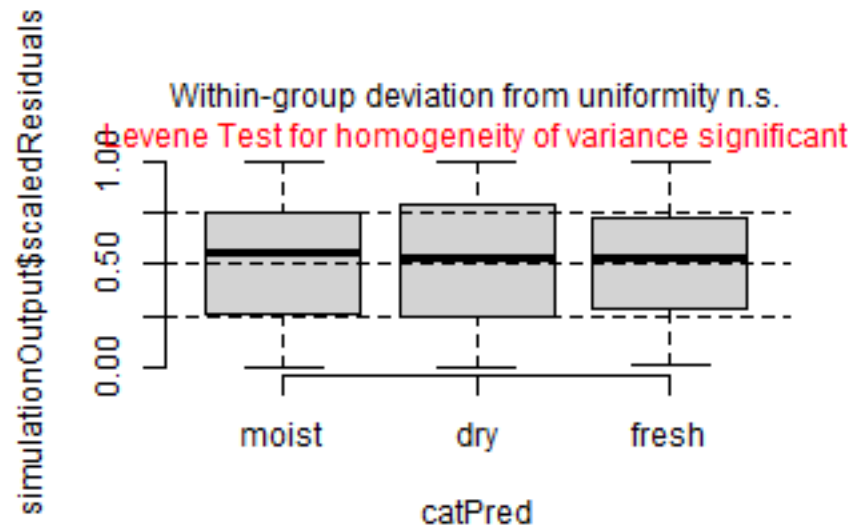
```
plotResiduals(simulation_output_2$scaledResiduals, sites$obs.year)
```



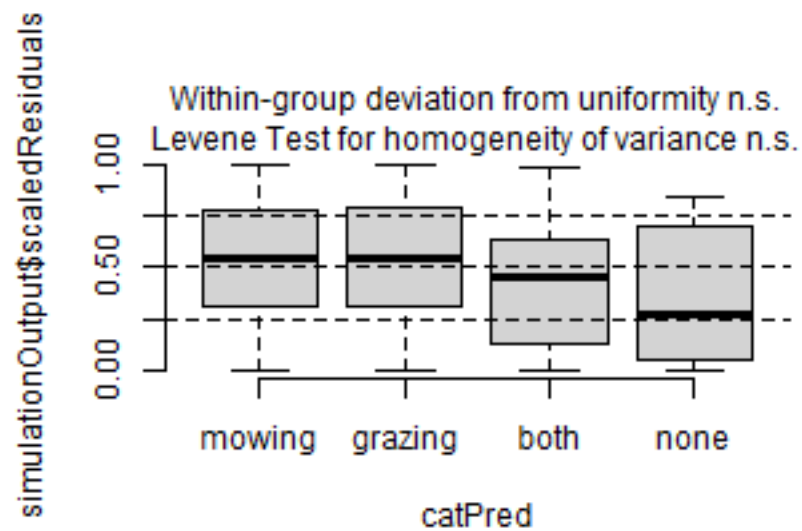
```
plotResiduals(simulation_output_1$scaledResiduals, sites$hydrology)
```



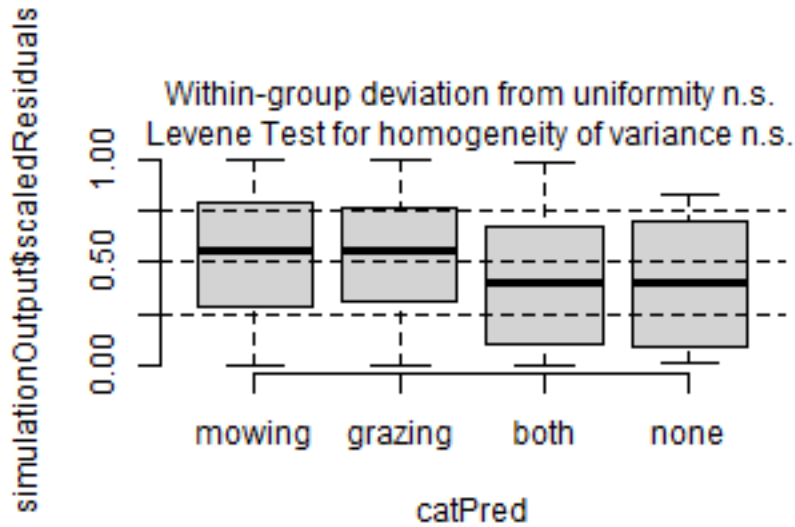
```
plotResiduals(simulation_output_2$scaledResiduals, sites$hydrology)
```



```
plotResiduals(simulation_output_1$scaledResiduals, sites$mngm.type)
```



```
plotResiduals(simulation_output_2$scaledResiduals, sites$mngm.type)
```



Check collinearity part 2 (Step 5)

Remove $VIF > 3$ or > 10 Zuur et al. 2010 Methods Ecol Evol DOI: 10.1111/j.2041-210X.2009.00001.x

```
car::vif(m_1)
```

```
##              GVIF Df  GVIF^(1/(2*Df))
## esy4          3.162398 1      1.778313
## site.type     1.810999 2      1.160058
## eco.id        1.637817 2      1.131270
## obs.year      1.042213 1      1.020888
## esy4:site.type 3.244952 2      1.342153
## esy4:eco.id    2.274876 1      1.508269
```

```
car::vif(m_2)
```

```
##              GVIF Df  GVIF^(1/(2*Df))
## esy4          5.478078 1      2.340529
## site.type     51.324896 2      2.676591
## eco.id        19.132558 2      2.091430
## obs.year      5.935369 1      2.436261
## esy4:site.type 8.242005 2      1.694370
## esy4:eco.id    2.578309 1      1.605711
## esy4:obs.year  3.245380 1      1.801494
## site.type:eco.id 68.507353 4      1.696161
## site.type:obs.year 9.514020 2      1.756269
## eco.id:obs.year 11.185396 2      1.828786
```

Model comparison

R2 values

```
MuMIn::r.squaredGLMM(m_1)
##              R2m      R2c
## [1,] 0.1307651 0.7185071
MuMIn::r.squaredGLMM(m_2)
```

```
##           R2m      R2c
## [1,] 0.185254 0.717501
```

AICc

Use AICc and not AIC since ratio $n/K < 40$ Burnahm & Anderson 2002 p. 66 ISBN: 978-0-387-95364-9

```
MuMin::AICc(m_1, m_2) %>%
  arrange(AICc)
##      df      AICc
## m_1 12 459.1683
## m_2 21 468.0383
```

Predicted values

Summary table

```
car::Anova(m_1, type = 3)

## Analysis of Deviance Table (Type III Wald chisquare tests)
##
## Response: log(y)
##              Chisq Df Pr(>Chisq)
## (Intercept)   1954.8888  1    < 2e-16 ***
## esy4           0.0043  1    0.94764
## site.type      0.1605  2    0.92290
## eco.id         12.9909  2    0.00151 **
## obs.year       2.3332  1    0.12664
## esy4:site.type  0.8595  2    0.65067
## esy4:eco.id     4.8966  1    0.02691 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

summary(m_1)

## Linear mixed model fit by maximum likelihood ['lmerMod']
## Formula: log(y) ~ esy4 * (site.type + eco.id) + obs.year + (1 | id.site)
## Data: sites
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##    458.0    502.1    -217.0    434.0      279
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.87810 -0.44449  0.01377  0.42939  2.60893
##
## Random effects:
## Groups Name Variance Std.Dev.
## id.site (Intercept) 0.2677  0.5174
## Residual          0.1282  0.3581
## Number of obs: 291, groups: id.site, 119
##
## Fixed effects:
##              Estimate Std. Error t value
## (Intercept)   -6.484504   0.146661 -44.214
```



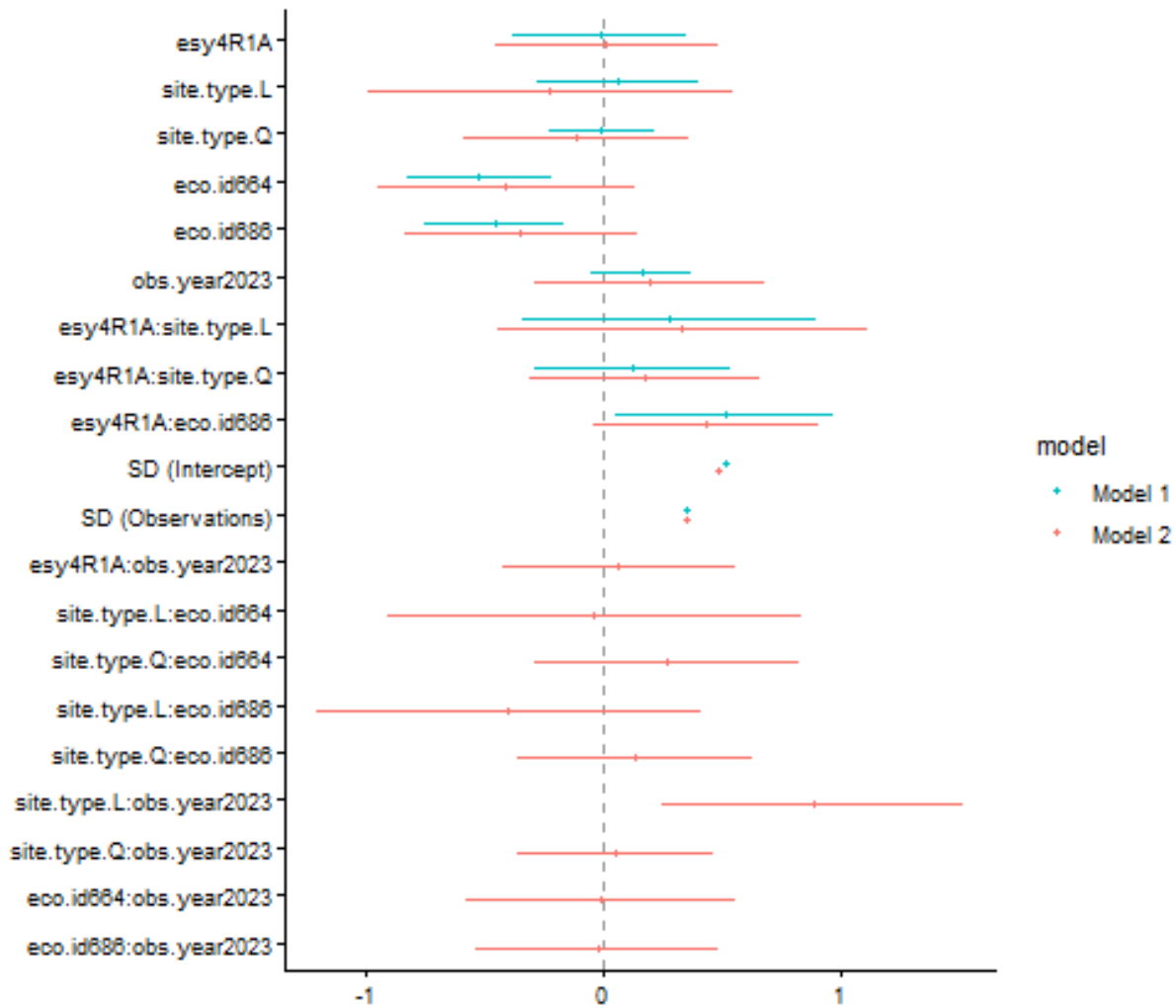
```
## esy4R1A          -0.012357    0.188179   -0.066
## site.type.L      0.066613    0.172951    0.385
## site.type.Q      -0.004452    0.113164   -0.039
## eco.id664        -0.520396    0.154064   -3.378
## eco.id686        -0.456739    0.150937   -3.026
## obs.year2023      0.164674    0.107807    1.527
## esy4R1A:site.type.L 0.284621    0.315409    0.902
## esy4R1A:site.type.Q 0.127330    0.211102    0.603
## esy4R1A:eco.id686  0.515454    0.232939    2.213
##
## Correlation of Fixed Effects:
##      (Intr) es4R1A st.t.L st.t.Q ec.664 ec.686 o.2023 e4R1A:...L
## esy4R1A      -0.610
## site.type.L  0.006 -0.076
## site.type.Q  0.440 -0.309  0.179
## eco.id664    -0.663  0.442  0.114 -0.051
## eco.id686    -0.632  0.439  0.050 -0.021  0.597
## obs.yer2023 -0.421  0.079  0.119 -0.024  0.077  0.007
## esy4R1A:...L -0.096  0.324 -0.484 -0.151 -0.012  0.011  0.016
## esy4R1A:...Q -0.220  0.497 -0.148 -0.466  0.034  0.027  0.004  0.459
## es4R1A:.686  0.340 -0.504 -0.047  0.009 -0.331 -0.503  0.019  0.166
##      e4R1A:...Q
## esy4R1A
## site.type.L
## site.type.Q
## eco.id664
## eco.id686
## obs.yer2023
## esy4R1A:...L
## esy4R1A:...Q
## es4R1A:.686  0.147
## fit warnings:
## fixed-effect model matrix is rank deficient so dropping 1 column / coefficient
```

Forest plot

```
dotwhisker::dwplot(
  list(m_1, m_2),
  ci = 0.95,
  show_intercept = FALSE,
  vline = geom_vline(xintercept = 0, colour = "grey60", linetype = 2)) +
  theme_classic()

## Package 'merDeriv' needs to be installed to compute confidence intervals
##   for random effect parameters.
## Package 'merDeriv' needs to be installed to compute confidence intervals
##   for random effect parameters.

## Warning: Using the `size` aesthetic with geom_segment was deprecated in ggplot2 3.4.0.
## i Please use the `linewidth` aesthetic instead.
## i The deprecated feature was likely used in the dotwhisker package.
##   Please report the issue at <https://github.com/fsolt/dotwhisker/issues>.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



Effect sizes

Effect sizes of chosen model just to get exact values of means etc. if necessary.

```
(emm <- emmeans(m_2, "esy4", type = "response"))
```

ESy EUNIS Habitat type

```
## esy4 response      SE  df lower.CL upper.CL
## R22  0.00125 0.000119 189  0.00104  0.00151
## R1A   nonEst    NA   NA      NA      NA
##
## Results are averaged over the levels of: site.type, eco.id, obs.year
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
## Intervals are back-transformed from the log scale
```

```
plot(emm, comparison = FALSE)
```

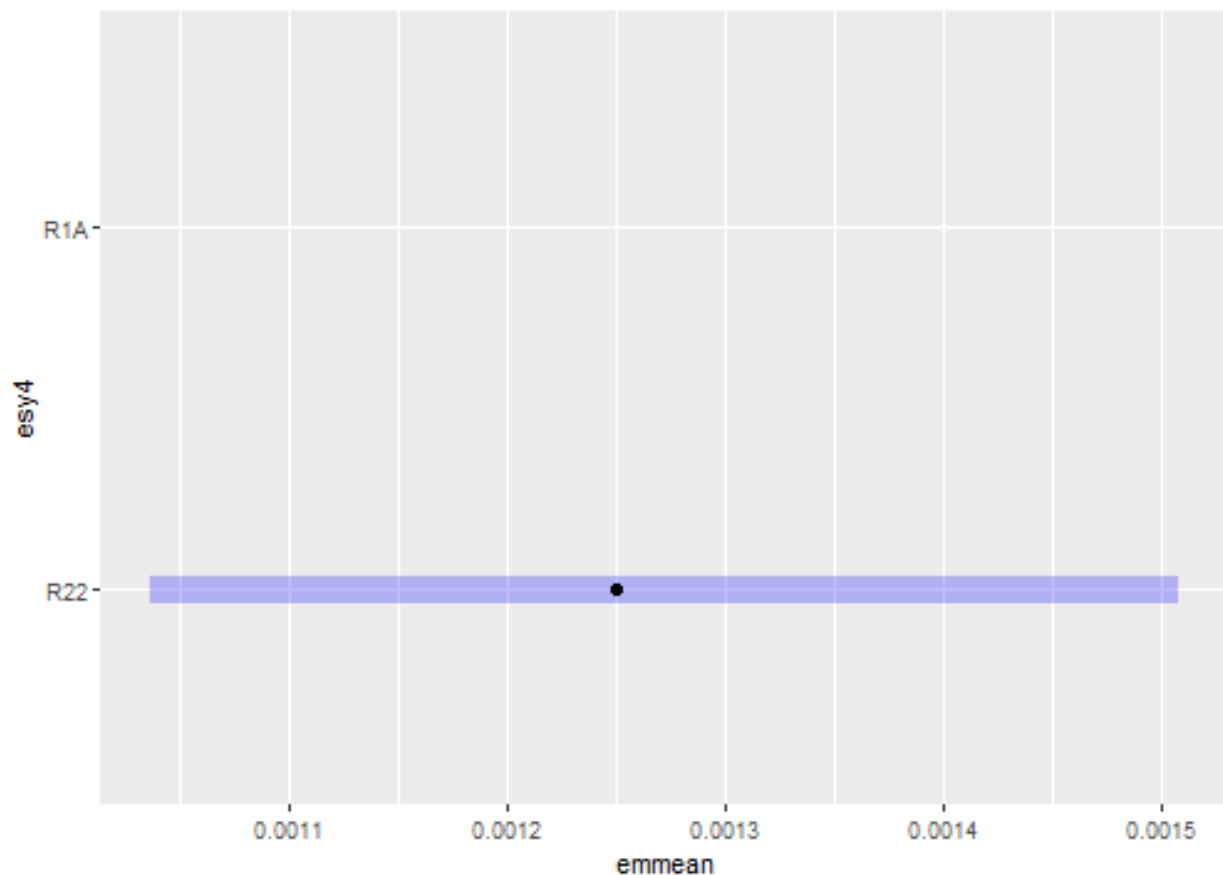
```
## Warning: `aes_()` was deprecated in ggplot2 3.0.0.
```

```
## i Please use tidy evaluation idioms with `aes()`
## i The deprecated feature was likely used in the emmeans package.
##   Please report the issue at <https://github.com/rvlenth/emmeans/issues>.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_segment()`).

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```



```
(emm <- emmeans(
  m_1,
  revpairwise ~ eco.id + esy4,
  type = "response"
))
```

Habitat type x Region

```
## $emmeans
## eco.id esy4 response      SE df lower.CL upper.CL
```

```

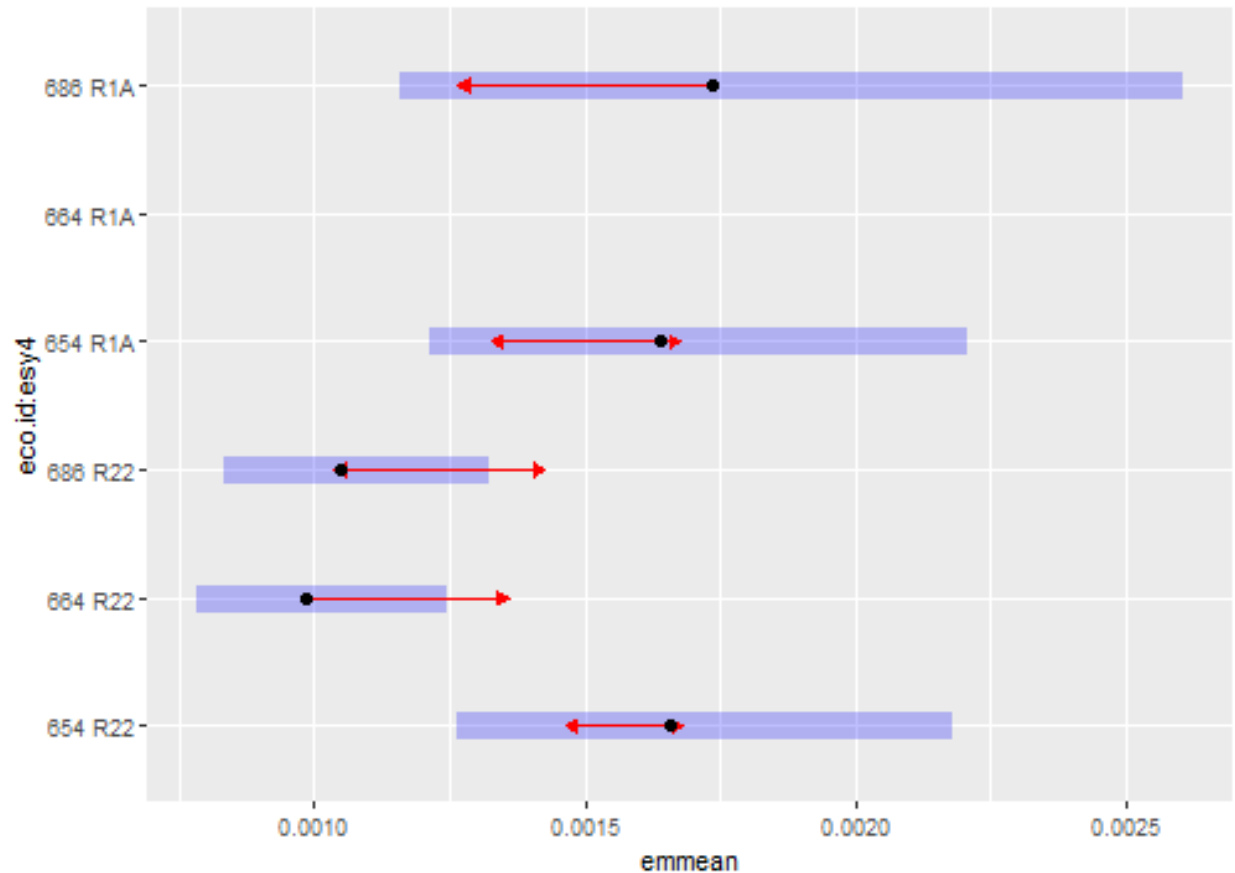
## 654 R22 0.001658 0.000229 163 0.001262 0.00218
## 664 R22 0.000985 0.000116 130 0.000781 0.00124
## 686 R22 0.001050 0.000123 157 0.000834 0.00132
## 654 R1A 0.001638 0.000247 140 0.001215 0.00221
## 664 R1A nonEst NA NA NA NA
## 686 R1A 0.001737 0.000357 183 0.001157 0.00261
##
## Results are averaged over the levels of: site.type, obs.year
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
## Intervals are back-transformed from the log scale
##
## $contrasts
## contrast ratio SE df null t.ratio p.value
## eco.id664 R22 / eco.id654 R22 0.594 0.0951 146 1 -3.251 0.0123
## eco.id686 R22 / eco.id654 R22 0.633 0.0992 158 1 -2.916 0.0326
## eco.id686 R22 / eco.id664 R22 1.066 0.1520 139 1 0.448 0.9916
## eco.id654 R1A / eco.id654 R22 0.988 0.1930 181 1 -0.063 1.0000
## eco.id654 R1A / eco.id664 R22 1.662 0.3160 138 1 2.669 0.0639
## eco.id654 R1A / eco.id686 R22 1.560 0.2950 150 1 2.347 0.1362
## eco.id664 R1A / eco.id654 R22 nonEst NA NA 1 NA NA
## eco.id664 R1A / eco.id664 R22 nonEst NA NA 1 NA NA
## eco.id664 R1A / eco.id686 R22 nonEst NA NA 1 NA NA
## eco.id664 R1A / eco.id654 R1A nonEst NA NA 1 NA NA
## eco.id686 R1A / eco.id654 R22 1.047 0.2610 172 1 0.186 0.9997
## eco.id686 R1A / eco.id664 R22 1.763 0.4170 169 1 2.397 0.1212
## eco.id686 R1A / eco.id686 R22 1.654 0.3650 226 1 2.281 0.1550
## eco.id686 R1A / eco.id654 R1A 1.060 0.2240 197 1 0.277 0.9987
## eco.id686 R1A / eco.id664 R1A nonEst NA NA 1 NA NA
##
## Results are averaged over the levels of: site.type, obs.year
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for comparing a family of 5 estimates
## Tests are performed on the log scale
plot(emm, comparison = TRUE)

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_segment()`).

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).

```



```
# (emm <- emmeans(
#   m_1,
#   reupairwise ~ site.type / esy4,
#   type = "response"
# ))
# plot(emm, comparison = TRUE)
```

Habitat type x Site type

Session info

```
## R version 4.5.2 (2025-10-31 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=German_Germany.utf8  LC_CTYPE=German_Germany.utf8
## [3] LC_MONETARY=German_Germany.utf8 LC_NUMERIC=C
## [5] LC_TIME=German_Germany.utf8
##
```

```

## time zone: Europe/Berlin
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## other attached packages:
## [1] emmeans_1.11.1  DHARMA_0.4.7    patchwork_1.3.0  ggbeeswarm_0.7.2
## [5] lubridate_1.9.4  forcats_1.0.0   stringr_1.5.2    dplyr_1.1.4
## [9] purrr_1.1.0     readr_2.1.5     tidyr_1.3.1      tibble_3.3.0
## [13] ggplot2_4.0.0   tidyverse_2.0.0 here_1.0.1
##
## loaded via a namespace (and not attached):
## [1] Rdpack_2.6.4      gridExtra_2.3      sandwich_3.1-1
## [4] rlang_1.1.6       magrittr_2.0.4     multcomp_1.4-28
## [7] compiler_4.5.2    mgcv_1.9-3         vctr_0.6.5
## [10] pkgconfig_2.0.3   crayon_1.5.3       fastmap_1.2.0
## [13] backports_1.5.0   labeling_0.4.3     utf8_1.2.6
## [16] ggstance_0.3.7    promises_1.3.2     rmarkdown_2.30
## [19] tzdb_0.5.0        nloptr_2.2.1       bit_4.6.0
## [22] xfun_0.53         later_1.4.2        broom_1.0.8
## [25] parallel_4.5.2    R6_2.6.1           gap.datasets_0.0.6
## [28] stringi_1.8.7     qgam_2.0.0         RColorBrewer_1.1-3
## [31] car_3.1-3         boot_1.3-32        estimability_1.5.1
## [34] Rcpp_1.1.0        iterators_1.0.14    knitr_1.50
## [37] zoo_1.8-14        parameters_0.25.0   httpuv_1.6.16
## [40] Matrix_1.7-4      splines_4.5.2       timechange_0.3.0
## [43] tidyselect_1.2.1  rstudioapi_0.17.1  abind_1.4-8
## [46] yaml_2.3.10       MuMIn_1.48.11       doParallel_1.0.17
## [49] codetools_0.2-20  lattice_0.22-7      plyr_1.8.9
## [52] bayestestR_0.15.3 shiny_1.10.0        withr_3.0.2
## [55] S7_0.2.0          evaluate_1.0.5      marginaleffects_0.25.1
## [58] survival_3.8-3    pillar_1.11.1       gap_1.6
## [61] carData_3.0-5     foreach_1.5.2       stats4_4.5.2
## [64] reformulas_0.4.1  insight_1.2.0       generics_0.1.4
## [67] vroom_1.6.6       rprojroot_2.0.4     hms_1.1.3
## [70] scales_1.4.0      minqa_1.2.8         xtable_1.8-4
## [73] glue_1.8.0        tools_4.5.2         data.table_1.17.8
## [76] lme4_1.1-37       mvtnorm_1.3-3       grid_4.5.2
## [79] rbibutils_2.3     datawizard_1.1.0    nlme_3.1-168
## [82] Rmisc_1.5.1       performance_0.13.0  beeswarm_0.4.0
## [85] vipor_0.4.7       Formula_1.2-5       cli_3.6.5
## [88] gtable_0.3.6      digest_0.6.37       pbkrtest_0.5.4
## [91] TH.data_1.1-3     farver_2.1.2        htmltools_0.5.8.1
## [94] lifecycle_1.0.4   mime_0.13           bit64_4.6.0-1
## [97] dotwhisker_0.8.4  MASS_7.3-65

```